Bounding Visual-Inertial Odometry Drift with Self-Supervised Aerial Geo-Localization for GNSS-Denied UAV Navigation

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***Abstract*—Visual-inertial odometry (VIO) is locally accurate but drifts without bound; image retrieval against satellite imagery is absolute but succeeds only intermittently. We combine the two on CPU for GNSS-denied navigation, using OpenVINS for motion and CAEVL — trained on reference imagery alone — for absolute position. On MUN-FRL `bell412_dataset4` (3.6 km, Ottawa International Airport) only 36% of retrievals are correct, and we show over two flights that the retrieval score is *negatively* correlated with correctness, so predictions cannot be gated individually. Rejecting outliers by mutual consensus through the motion prior instead raises precision from 17.7% to 99.0% while keeping 93% of correct predictions. A causal similarity-transform estimator then reduces position RMSE from 489 ± 44 m to 248 ± 86 m (49%, three runs) using only information available at each instant. We also detect and automatically correct a 456 m georeferencing error in the dataset's shipped map metadata, and identify the binding limitation: retrieval is physically possible only during the 34% of this flight above 66 m, since below that the camera cannot image a full map tile.**

I. Introduction

A UAV without GNSS must localise from onboard sensors. VIO integrates motion, so its error grows without limit — here OpenVINS [1] drifts to 700 m over 3.6 km, 19% of distance flown. Matching the downward camera against georeferenced satellite imagery gives absolute position that never drifts, but fails often: terrain must be distinctive, the aircraft high enough to image useful ground, and the reference imagery is months or years old.

The two are complementary, which motivates fusion. The difficulty — and the substance of this work — is that the retrieval output is dominated by outliers and its confidence score cannot identify them. The contribution is therefore about *which measurements to believe*, using the motion prior rather than appearance.

We use CAEVL [2], which trains on reference imagery alone: no paired UAV-satellite examples, no correspondences. It operates on Canny edge maps [3] to suppress the domain gap, via an autoencoder with a DINOv2 [4] perceptual loss followed by VICReg [5] fine-tuning on synthetically degraded copies. Deploying over new terrain needs only satellite tiles.

Everything runs on CPU: OpenVINS at 20 Hz, retrieval at 4.3 Hz.

II. Dataset Selection

The system needs a downward camera, a synchronised IMU, metric ground truth and enough distance for drift to matter. Few datasets provide all four (Table I).

TABLE I
Only MUN-FRL provides raw IMU with a nadir camera and metric truth.

Dataset	Altitude	IMU	Outcome
VPAIR [6]	300 m	no	sparse frames
AerialVL [7]	—	no	camera + GNSS only
SatLoc	100–300 m	not released	unusable
ALTO	257 m	no	straight line
Zurich Urban MAV	5–15 m	yes	too low
MUN-FRL [8]	**25–170 m**	**yes**	**selected**

The scarcity is structural: VIO needs parallax, which favours low flight; retrieval needs ground footprint, which favours high. Dataset authors optimise for one. This tension reappears below as the dominant limitation.

We use `bell412_dataset4`: 3.6 km over 319 s from a Bell 412 over Ottawa International Airport — nadir FLIR camera (1440×1080, 20 Hz), Xsens MTi-30 IMU at 400 Hz, hardware-synchronised, RTK ground truth at 5 Hz.

III. Data Preparation

A. IMU Timestamps

Recorded IMU timestamps are driver *receipt* times: mean interval 2500.9 µs (correct for 400 Hz) but standard deviation 2538 µs, with gaps from 25 µs to 19 ms. Since VIO integrates over time, this corrupts propagation. The dataset publishes the sensor's hardware clock on a separate topic in 1:1 correspondence; fitting an affine map and restamping reduces the interval standard deviation to 68.5 µs, a factor of 37, with no samples dropped.

B. A 456 m Georeferencing Error

MUN-FRL encodes each map's georeferencing in its filename. Projecting the truth track with those parameters places it **456 m** from the actual flight (Fig. 1). A bounds check cannot catch this — the track still lands inside the image — yet every tile then depicts different ground than it claims. Before

this was found, query and reference tiles nominally 6 m apart showed forest canopy and ploughed farmland respectively.

The authors drew the true path on each map in green, permitting automatic recovery: we fit the scale and offset placing the known track onto that curve, scoring by mean distance under a distance transform. Isolation is exact, since the imagery is monochrome and only the annotation carries colour. The fit converges to **0.35 px (0.28 m)**, confirming the published scale is right and only the offsets are wrong — by 378 m east and 256 m north.

C. Reference Imagery

The shipped maps remain unusable even once corrected: the painted path is 18.6 m wide and falls inside nearly every tile a query could match, and at 0.79 m/px they are coarser than the queries. Inpainting removes the false edge but smears 23% of each tile.

We instead fetch Esri World Imagery [9] at zoom 19 (0.21 m/px), 2116 tiles over 6.75 km^2 (Fig. 2). The ENU-to-pixel mapping is solved by least squares against the PPK latitudes and longitudes, residual **0.30 px (0.06 m)**. As an independent check, the PPK solution leads the bag by exactly 18.00 s — the 2022 GPS-UTC leap second count.

D. Query Generation

Tiles cover fixed ground; a frame covers area proportional to altitude. Each output pixel is treated as a ground-plane point in ENU, rotated through the full VIO attitude and the Kalibr [10] extrinsic, then projected through the equidistant model:

$$\boldsymbol{P}_{\text{enu}} = (\text{east},\ \text{north},\ z_{\text{ground}} - z_{\text{cam}}) \tag{1}$$

$$\boldsymbol{P}_{\text{cam}} = \boldsymbol{R}_{C\to I}^{\top}(\boldsymbol{R}_{G\to I}\boldsymbol{P}_{\text{enu}} - \boldsymbol{p}_{C\ \text{in}\ I}) \tag{2}$$

$$\boldsymbol{u} = \pi_{\text{equi}}(\boldsymbol{P}_{\text{cam},xy}/\boldsymbol{P}_{\text{cam},z}) \tag{3}$$

Full attitude matters: the aircraft banks, and this sequence sweeps 11 422° of heading. An earlier version building the grid in camera axes — leaving queries rotated while tiles are north-up — gave retrieval indistinguishable from chance. Attitude must come from the VIO, since the PPK solution is position-only.

IV. Method

A. VIO Characterisation

OpenVINS runs monocular on the nadir camera with online calibration disabled (the dataset supplies Kalibr values, and this trajectory does not excite every calibration state). The camera–IMU time offset is left free and converges to -28.5 ms against the dataset's published -29.7 ms, an independent check that the configuration is right.

Anchored on the first 20 s — what a real vehicle has, knowing its takeoff pose — the estimate starts 14.7 m from truth and drifts to 700 m (Fig. 3). A best-fit similarity reduces this to 215 m: most of the error is a systematic **40% scale deficit**, not random walk. The cause was not identified, and survived IMU restamping.

B. The Retrieval Score Cannot Gate

Over both flights, retrieval similarity is *negatively* correlated with correctness: 0.9429 for correct predictions against 0.9493 for incorrect on `dataset4`, and 0.9447 against 0.9598 on `dataset3`. The tile-size ablation below explains the mechanism — descriptor collapse makes confidence highest exactly where tiles are least distinguishable. No threshold on the score can separate inliers; rejection must be geometric.

C. Consensus Rejection

Incorrect predictions are near-uniform over the database, so two independent errors agreeing within τ — after removing the VIO displacement between their timestamps — occurs with probability of order $\pi\tau^2/A$, about 0.5% here ($N = 1612$, $A \approx 1.1\,\text{km}^2$, $\tau = 40\,\text{m}$). Correct predictions agree by construction. Requiring m mutually consistent predictions suppresses outliers as $(\pi\tau^2/A)^{m-1}$ while retaining inliers.

Algorithm 1 Consensus outlier rejection

Require: predictions $\{(t_i, \boldsymbol{f}_i)\}$, VIO $\boldsymbol{v}(t)$, window W, threshold τ, support m

$\mathcal{K} \leftarrow \emptyset$
for all windows $\mathcal{W}$ of width W **do**
  $best \leftarrow \emptyset$
  for n iterations **do**
    sample $i, j \in \mathcal{W}$; $s \leftarrow \|\boldsymbol{f}_j - \boldsymbol{f}_i\|/\|\boldsymbol{v}_j - \boldsymbol{v}_i\|$
    $\theta \leftarrow \angle(\boldsymbol{f}_j - \boldsymbol{f}_i) - \angle(\boldsymbol{v}_j - \boldsymbol{v}_i)$; $\boldsymbol{t} \leftarrow \boldsymbol{f}_i - s\boldsymbol{R}(\theta)\boldsymbol{v}_i$
    $\mathcal{I} \leftarrow \{k : \|s\boldsymbol{R}(\theta)\boldsymbol{v}_k + \boldsymbol{t} - \boldsymbol{f}_k\| < \tau\}$
    if $|\mathcal{I}| > |best|$ **then**
      $best \leftarrow \mathcal{I}$
    end if
  end for
  if $|best| \geq m$ **then**
    $\mathcal{K} \leftarrow \mathcal{K} \cup best$
  end if
end for
return $\mathcal{K}$

D. Causal Fusion

The VIO is mapped into the map frame by a similarity transform estimated online:

$$\hat{\boldsymbol{p}}_k = s\,\boldsymbol{R}(\theta)\,\boldsymbol{p}_k^{\text{vio}} + \boldsymbol{t} \tag{4}$$

Estimating s and θ is essential: OpenVINS initialises with arbitrary heading, and its scale deficit would otherwise reappear between every pair of predictions.

The estimator is strictly causal. Predictions enter a buffer as their timestamps elapse; a *trailing* window is screened for consensus; survivors accumulate; the transform is refitted by Umeyama's method [11] to all accepted history. Before the transform is available the estimator dead-reckons on the VIO from the known takeoff pose — a real vehicle has this from a survey or the last GNSS fix — so a position is output at every instant. Handover occurs at $t + 73$ s.

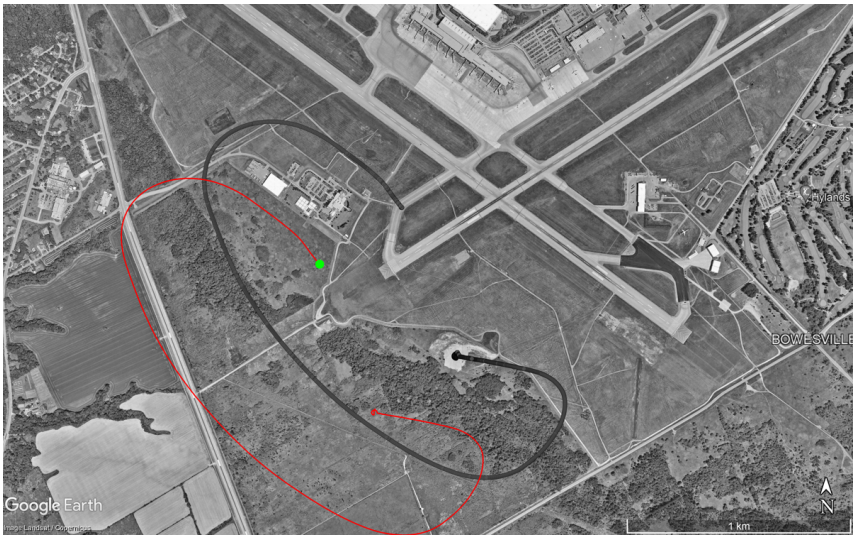


Fig. 1. Red: the track projected with the published parameters. Dark curve: the true path, drawn on the map by the dataset authors. The two differ by 456 m.

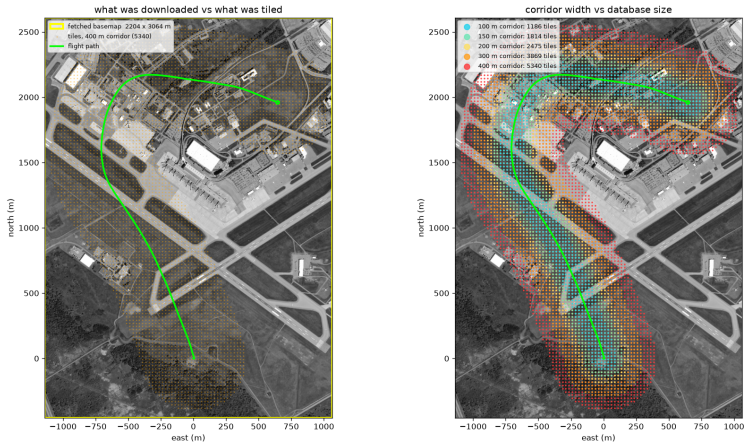


Fig. 2. Left: fetched basemap (yellow) against the tiles cut from it. Right: database size versus corridor width, 5340 tiles at 400 m down to 1186 at 100 m.

An EKF over (x, y, s, θ) with chi-square gating was tried first and performs worse (422 m against 347 m). Two reasons. *Gate lockout*: a scale error makes position error grow linearly while random-walk noise grows as $\sqrt{\cdot}$, so the filter becomes overconfident and rejects correct predictions — most certain when most wrong (Fig. 6). *Weak observability*: a per-prediction position update lets position absorb the innovation, leaving s to converge too slowly.

V. Results

A. Retrieval

Restricting the database to a corridor around positions where queries exist — in deployment, the motion prior's uncertainty region — halves the tile count and doubles recall (Table II, Fig. 7). 83% of erroneous predictions matched tiles that could never have been correct.

Reducing tile size to 60 m, to admit lower-altitude frames, *degrades* performance: at 60 m, 2.21% of tile pairs exceed 0.99 similarity against 0.31% at 100 m. Descriptors collapse, which also explains the anti-correlated confidence score. Runtime is flat across a 3.3× database range, confirming cost is query encoding rather than search.

TABLE II
Retrieval on bell412_dataset4.

Configuration	Tiles	R@100 m	Median	ms
100 m tile, 400 m corridor	5340	17.7%	406 m	229
100 m tile, 200 m corridor	1612	**35.8%**	**230 m**	233
60 m tile, 200 m corridor	3270	13.6%	476 m	235

A residual error class survives any corridor: 52 incorrect predictions lie within 100 m of the path yet carry 398 m median error, with a worst case of 2036 m. These are along-track confusions — a corridor constrains distance to the path as a curve, not position along it — and must be removed by the temporal consensus test instead.

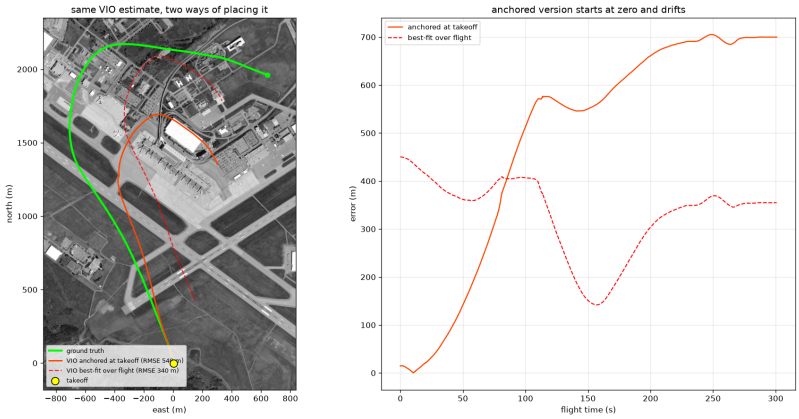


Fig. 3. Anchored at takeoff (orange), OpenVINS starts on the truth and drifts to 700 m. A best-fit placement (dashed) distributes the same error differently.

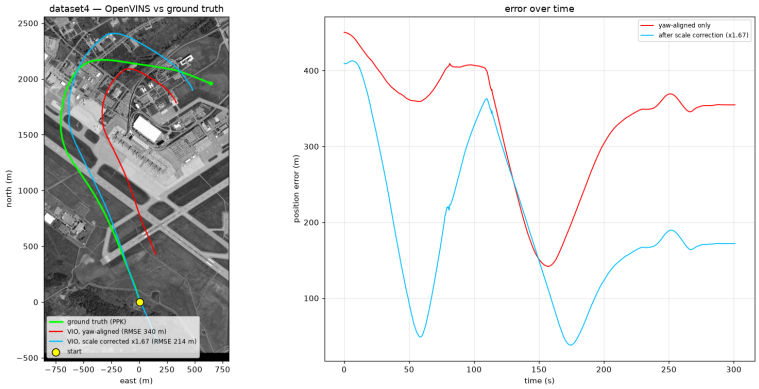


Fig. 4. The yaw-aligned estimate (red) is a shrunken copy of truth; scaling by 1.67 (blue) recovers the shape. Even then the error oscillates between 35 and 360 m, indicating orientation error alongside scale.

B. Terrain Governs Performance

The same pipeline, aircraft, airport and weights give 8.9% R@100 m on `dataset3` against 17.7% on `dataset4`. `dataset3` loops over continuous forest; `dataset4` crosses the terminal apron and taxiways (Figs. 8–11). Over uniform canopy, Canny edge maps are texture noise and every tile resembles every other. Altitude does not explain it (Fig. 10).

C. Fusion

TABLE III
FUSION, BOTH TRAJECTORIES ANCHORED AT TAKEOFF. CAUSAL USES ONLY INFORMATION AVAILABLE AT EACH INSTANT.

	OpenVINS	Fused	Reduction
RMSE (3 runs)	489 ± 44 m	$\mathbf{248 \pm 86}$ **m**	49%
best run	463.8 m	**198.2 m**	57%
Trajectory drift	$17.7 \pm 1.5\%$	$\mathbf{10.1 \pm 3.6}\%$	43%
Offline, cruise	166 m	**48 m**	71%

D. Run-to-run Variance

OpenVINS was run three times on identical inputs. Its dynamic initialiser starts from assumed zero bias and solves a nonlinear problem over the first data window, so the outcome is not deterministic (Table IV, Fig. 14). Runs 2 and 3 agree to within 0.1 m and 0.002 in scale; run 1 is a clear outlier — a small number of distinct initialisation basins rather than continuous noise.

The variance propagates more than proportionally: run 3's *fused* error is 43% below run 1's while its *VIO* error is only 14% below, because the consensus test uses VIO motion to judge agreement. Better VIO gives cleaner consensus gives a better transform.

TABLE IV
THREE OPENVINS RUNS, IDENTICAL CONFIGURATION AND DATA.

	VIO RMSE	Fused RMSE	Reduction	Scale
run 1	540.5 m	346.8 m	36%	1.522
run 2	463.7 m	199.0 m	57%	1.299
run 3	463.8 m	198.2 m	57%	1.297
mean	489.3	248.0	49%	1.373
s.d.	44.3	85.6	—	0.130

E. Computational Cost

OpenVINS is comfortably real-time: 14.1 ms per frame against a 50 ms budget, 60% of one core, 315 MB. Retrieval

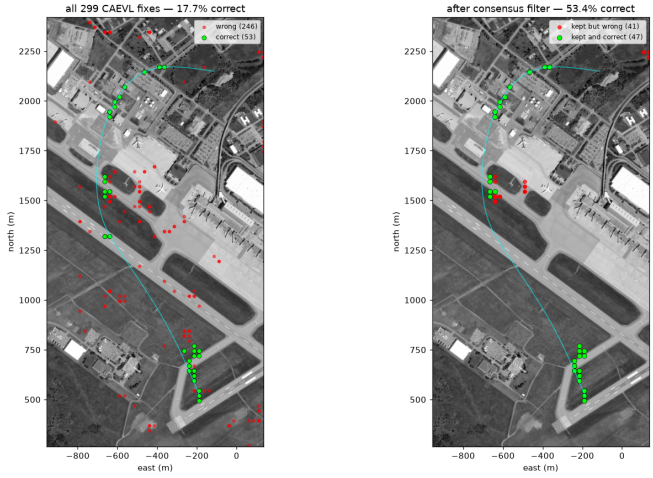


Fig. 5. Left: all predictions, green correct. Right: after consensus. Precision rises from 17.7% to 99.0% while 93% of correct predictions survive.

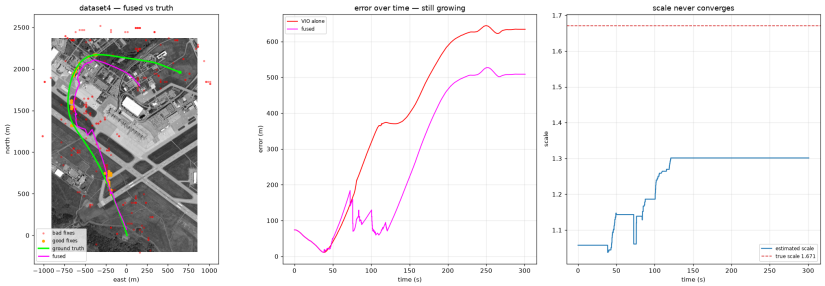


Fig. 6. The EKF failing. Accepted predictions (orange) cluster early; the scale estimate advances for 120 s then freezes permanently as the gate closes.

is the expensive half — 233 ms per query at 4.3 Hz, but that depends on 20 threads. At 4.4 CPU-seconds per query, a single core gives roughly 0.23 Hz. Since fixes are consumed intermittently that may suffice, but the multi-core figure should not be quoted alone for an embedded computer.

VI. Limitations

Altitude coverage binds. A 100 m tile needs ∼66 m of altitude, since the camera sees about 1.5× its height. Only 34% of this flight qualifies: takeoff, a 70 s cruise near 105 m, descent, then 80 s at ∼11 m where the camera images 17 m of ground. Running retrieval on those frames anyway was tested and made fusion *worse*, 347 to 527 m — 260 predictions below 66 m were 1.9% correct, and consensus precision fell from 87% to 43%. Reducing tile size to compensate fails for the reasons above.

Two quantities come from ground truth: query altitude and the VIO-to-map yaw. In deployment these come from an altimeter and a magnetometer; MUN-FRL records neither. The VIO's own altitude is not a substitute — median error 29.8 m, a 30% error in every query's ground extent at 100 m.

Fusion is evaluated on one flight, retrieval on two.

The 40% VIO scale deficit is unexplained, and is also the source of the run-to-run spread.

VII. Conclusion

A CPU-only pipeline bounds VIO drift using self-supervised geo-localization requiring no paired training data, on real flight data with centimetre-accurate truth: 489 ± 44 m to 248 ± 86 m in real time, and 166 to 48 m offline where fixes exist.

The principal finding is methodological. With inlier rates below 20% and a confidence score anti-correlated with correctness, deciding *which* measurements to accept matters more than the estimator consuming them. Consensus among temporally adjacent predictions, judged through the motion prior, raises precision from 17.7% to 99.0% while keeping 93% of correct predictions — without consulting the retrieval score or the filter's own belief, structurally avoiding the divergence mode in which a drifted filter rejects the measurements that would correct it.

The limiting factor is availability, not accuracy. Query-side scale normalisation [12], building one database at the lowest operating altitude and rescaling queries down, would remove both the altitude dependence and the fixed-footprint assumption. Retraining CAEVL at a tile footprint matched to deployment altitude — rather than the 14.4× larger ViLD footprint — is a training run rather than a research problem, and is the change most likely to move the 35.8% retrieval figure.

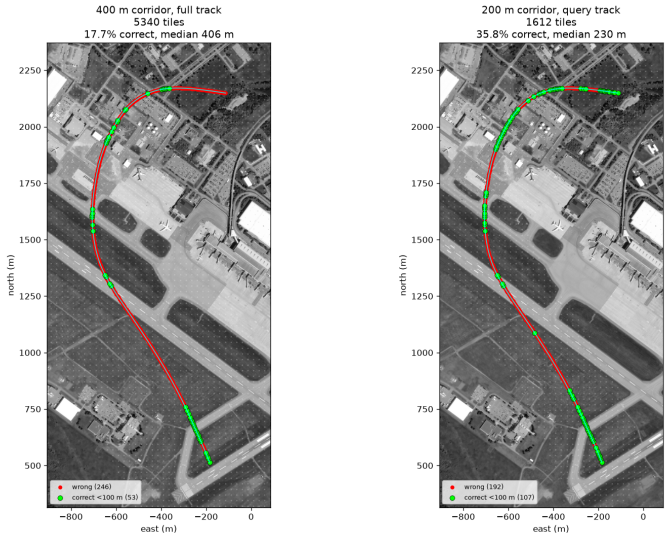


Fig. 7. Restricting the database from 5340 to 1612 tiles doubles recall, 17.7% to 35.8%.



Fig. 8. `dataset3` (left) loops over forest; `dataset4` (right) crosses the terminal apron and taxiways.

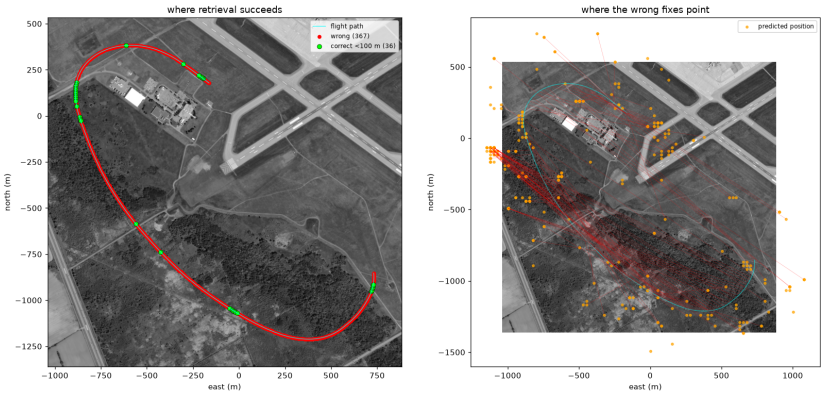


Fig. 9. `dataset3`: successes cluster near the airport buildings; the southern arc over forest fails almost entirely.

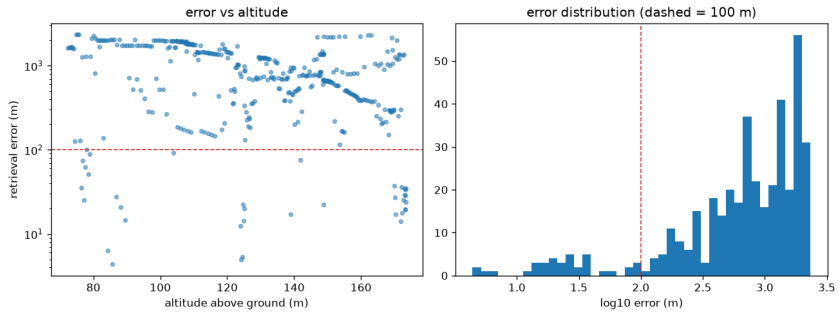


Fig. 10. Error against altitude on `dataset3`. No monotonic relationship, which first indicated that scene content rather than scale governs performance.

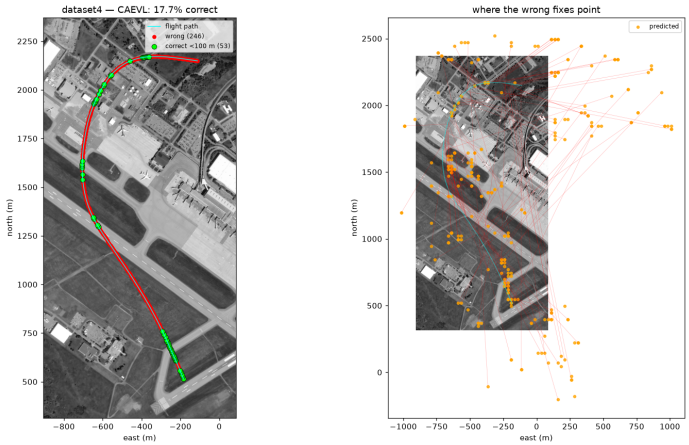


Fig. 11. `dataset4`: successes concentrate on runway markings and the terminal apron; the open stretch between runways fails.

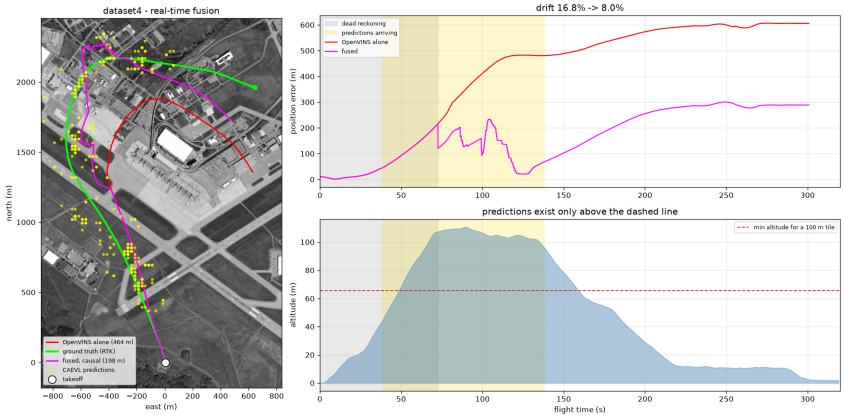


Fig. 12. Causal fusion. Red: OpenVINS. Green: RTK truth. Magenta: fused. Grey: dead reckoning before handover. Gold: predictions arriving. Bottom right: the online scale estimate, held at unity during dead reckoning then learned, though it does not fully converge before predictions cease.

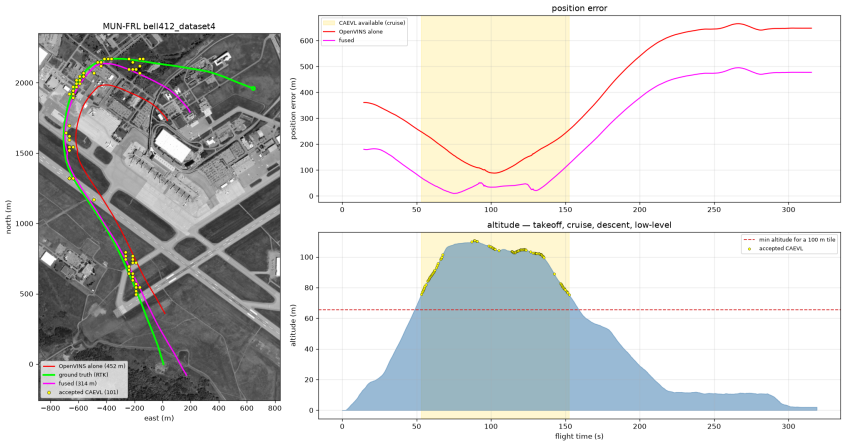


Fig. 13. Offline fusion with the altitude profile. Predictions coincide exactly with the portion of flight above the 65.6 m needed to image a full tile.

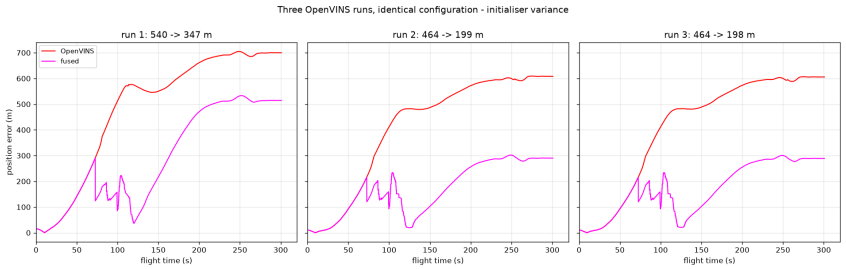


Fig. 14. Three OpenVINS runs on identical inputs. Runs 2 and 3 are nearly indistinguishable; run 1 initialised into a worse basin.

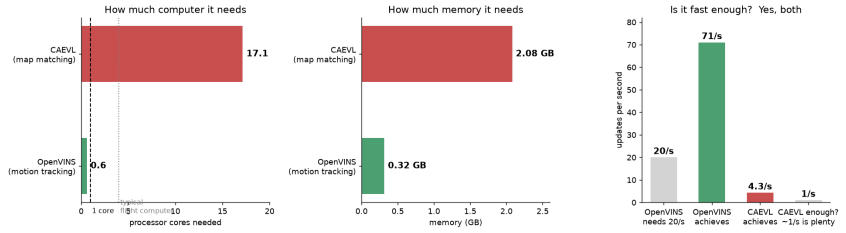


Fig. 15. Measured cost. Motion tracking needs under one core and runs 3.5× faster than required; map matching needs about seventeen.

References

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